```
0 -m limit --limit 1/min -j LOG –log-prefix "blacklisted"
# firewall-cmd –direct –permanent –add-rule ipv4 raw blacklist
1 -j DROP
```

Rich rules: Firewalld rich rules give administrators an expressive language with which to express custom firewall rules that are not covered by basic firewalld syntax — for example, to only allow connections to a service from a single IP address instead of all IP addresses routed through a zone.

Rich rules can be used to express basic allow/deny rules, and can also be used to configure logging to rsyslog and auditd as well as port forwards, masquerading and rate limiting.

The basic syntax of a rich rule can be expressed by the following block**:**

```
rule
    [source]
    [destination]
    service|port|protocol|icmp-block|masquerade|forward-port
    [log]
    [audit]
    [accept|reject|drop]
```

Almost every single element of a rule can take additional arguments in the form of *option=value.*

Note: For the full syntax of rich rules, consult the *firewalld.richlanguage* man page in Red Hat Enterprise Linux 7.

Rule ordering

Once multiple rules have been added to a zone (or the firewall, in general), the ordering of rules can have a big effect on how the firewall behaves.

The basic ordering of rules inside a zone is the same for all zones:

- Port forwarding and masquerading rules set for that zone.
- Logging rules set for that zone.
- Allow rules set for that zone
- Deny rules set for that zone.

In all cases, the first match will win. If a packet has not been matched by any rule in a zone, it is typically denied, but zones might have different defaults — for example, the trusted zone will accept any unmatched packet. Also, after matching a logging rule, a packet will continue to be processed as normal.

Direct rules are an exception. Most direct rules are parsed before any other processing is done by firewalld, but the direct syntax allows an administrator to insert any desired rule anywhere in any zone.

Testing and debugging

To make testing and debugging easier, almost all rules can be added to the runtime configuration with a timeout. The moment the rule with a timeout is added to the firewall, the timer starts counting down for that rule. Once the timer for that rule reaches zero seconds, that rule is removed from the runtime configuration.

Using timeouts can be an incredibly useful tool while working on remote firewalls, especially when testing more complicated rule sets. If a rule works, the administrator can add it again but with the *–permanent* option (or at least with a timeout). If the rule does not work as intended, maybe even locking the administrator out of the system, it will be removed automatically, allowing the administrator to continue his or her work.

A timeout is added to a runtime rule by adding the option *--timeout=<TIMEINSECONDS>* to the end of *firewall-cmd* that enables the rule.

Working with rich rules

Firewall-cmd has four options for working with rich rules. All of these options can be used in combination with the regular *–permanent* or *–zone=<ZONE>* options.

Some options with explanations are given below.

- *--add-rich-rule='<RULE>':* Adds *<RULE>* to the specified zone, or the default zone if no zone is specified.
- *--remove-rich-rule='<RULE>':* Removes *<RULE>* from the specified zone, or the default zone if no zone is specified.
- *--query-rich-rule='<RULE>':* Queries if *<RULE>*has been added to a specified zone, or the default zone if no zone is specified. Returns 0 if the rule is present, otherwise returns 1.
- *--list-rich-rules:* Outputs all rich rules for the specified zone or the default zone if no zone is specified.

All configured rich rules are also shown in the output from *firewall-cmd –list-all* and *firewall-cmd –list-all-zones*.

Rich rule examples

Shown below are some examples of rich rules.

- *# firewall-cmd –permanent –zone=classroom –add-rich-rule='rulefamily=ipv4 source address=192.168.0.11/32 reject'*

This rule is to reject all traffic from the IP address 192.168.0.11 in the classroom zone.

When using the source or destination with an address option, the *family=option* rule must be set to either IPv4 or IPv6.

- *# firewall-cmd –add-rich-rule='rule service name=ftp limit value=2/m accept'*

This rule is to allow two new connections to FTP per minute in the default zone. This change is only made in runtime configuration.

- *#firewall-cmd –permanent –add-rich-rule='rule protocol value=espdrop'*

This drops all incoming IPsec ESP protocol packets from anywhere in the default zone.